

Unit 1: The Real Number System — ANSWER KEY

NC.8.NS.1 · NC.8.NS.2

Part A — Classify: Rational or Irrational

1. 0.333... (repeating)
(A) Rational (B) Irrational
✓ **Answer: Rational**
Repeating decimal = rational
2. π (3.14159...)
(A) Rational (B) Irrational
✓ **Answer: Irrational**
Non-repeating, non-terminating
3. $\sqrt{49}$
(A) Rational (B) Irrational
✓ **Answer: Rational**
 $\sqrt{49} = 7$
4. $\sqrt{10}$
(A) Rational (B) Irrational
✓ **Answer: Irrational**
Not a perfect square
5. $-\frac{3}{8}$
(A) Rational (B) Irrational
✓ **Answer: Rational**
Fraction = rational
6. 0.121212... (repeating)
(A) Rational (B) Irrational
✓ **Answer: Rational**
Repeating decimal = rational
7. $\sqrt{144}$
(A) Rational (B) Irrational
✓ **Answer: Rational**
 $\sqrt{144} = 12$
8. 0.101001000100001...
(A) Rational (B) Irrational
✓ **Answer: Irrational**
Pattern but NOT repeating
9. $\frac{2}{7}$

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(A) Rational

(B) Irrational

✓ **Answer: Rational**

$2/7 = 0.285714\dots$ repeats

10. $\sqrt{2}$

(A) Rational

(B) Irrational

✓ **Answer: Irrational**

Not a perfect square

11. $\sqrt[3]{64}$

(A) Rational

(B) Irrational

✓ **Answer: Rational**

$\sqrt[3]{64} = 4$

12. $\sqrt[3]{30}$

(A) Rational

(B) Irrational

✓ **Answer: Irrational**

30 is not a perfect cube

Part B — Convert Between Fractions & Decimals

1. Write $5/8$ as a decimal:

(A) 0.625

(B) 0.85

(C) 0.58

(D) 1.6

✓ **Answer: (A) 0.625**

$5 \div 8 = 0.625$

2. Write 0.6 as a fraction (simplest form):

(A) $6/10$

(B) $3/5$

(C) $2/3$

(D) $1/6$

✓ **Answer: (B) $3/5$**

$0.6 = 6/10 = 3/5$

3. Write $0.4444\dots$ as a fraction (simplest form):

(A) $4/10$

(B) $2/5$

(C) $4/9$

(D) $1/4$

✓ **Answer: (C) $4/9$**

one repeating digit $\rightarrow /9 = 4/9$

4. Write $0.7272\dots$ as a fraction (simplest form):

(A) $72/100$

(B) $8/11$

(C) $7/9$

(D) $18/25$

✓ **Answer: (B) $8/11$**

two digits $\rightarrow 72/99 = 8/11$

5. Write $7/20$ as a decimal:

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- (A) 0.35 (B) 0.72 (C) 0.720 (D) 0.035

✓ **Answer: (A) 0.35**

$$7 \div 20 = 0.35$$

6. Write 0.125 as a fraction (simplest form):

- (A) $1/4$ (B) $1/5$ (C) $1/8$ (D) $5/8$

✓ **Answer: (C) $1/8$**

$$0.125 = 125/1000 = 1/8$$

Part C — Estimate Each Square or Cube Root

1. $\sqrt{5}$ is closest to:

- (A) 2.2 (B) 2.0 (C) 2.5 (D) 2.6

✓ **Answer: (A) 2.2**

$$\sqrt{5} \approx 2.24$$

2. $\sqrt{10}$ is closest to:

- (A) 3.0 (B) 3.2 (C) 3.6 (D) 3.5

✓ **Answer: (B) 3.2**

$$\sqrt{10} \approx 3.16$$

3. $\sqrt{20}$ is closest to:

- (A) 4.0 (B) 4.5 (C) 5.0 (D) 4.2

✓ **Answer: (B) 4.5**

$$\sqrt{20} \approx 4.47$$

4. $\sqrt{50}$ is closest to:

- (A) 7.0 (B) 7.1 (C) 7.5 (D) 6.5

✓ **Answer: (B) 7.1**

$$\sqrt{50} \approx 7.07$$

5. $\sqrt{8}$ is closest to:

- (A) 2.5 (B) 2.8 (C) 3.0 (D) 2.6

✓ **Answer: (B) 2.8**

$$\sqrt{8} \approx 2.83$$

6. $\sqrt{40}$ is closest to:

- (A) 6.0 (B) 6.3 (C) 6.5 (D) 7.0

✓ **Answer: (B) 6.3**

$$\sqrt{40} \approx 6.32$$

7. $\sqrt[3]{30}$ is closest to:

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(A) 3.1

(B) 3.9

(C) 5.5

(D) 2.7

✓ **Answer: (A) 3.1**

$$27 < 30 < 64 \rightarrow \sqrt[3]{30} \approx 3.1$$

8. $-\sqrt{40}$ is closest to:

(A) 6.3

(B) -6.3 (C) -5.5 (D) -20 ✓ **Answer: (B) -6.3**

$$\sqrt{40} \approx 6.3, \text{ so } -\sqrt{40} \approx -6.3$$

Part D — Order from Least to Greatest**1.** 3.5, $\sqrt{10}$, $10/3$, π ✓ **Answer: π , $\sqrt{10}$, $10/3$, 3.5**

$$\pi \approx 3.14, \sqrt{10} \approx 3.16, 10/3 \approx 3.33, 3.5$$

2. 1.5, $\sqrt{2}$, $7/4$, $\sqrt{3}$ ✓ **Answer: $\sqrt{2}$, 1.5, $\sqrt{3}$, $7/4$**

$$\sqrt{2} \approx 1.41, 1.5, \sqrt{3} \approx 1.73, 7/4 = 1.75$$

3. $\sqrt{5}$, 2.5, $11/5$, $\sqrt{7}$ ✓ **Answer: $11/5$, $\sqrt{5}$, 2.5, $\sqrt{7}$**

$$11/5 = 2.2, \sqrt{5} \approx 2.24, 2.5, \sqrt{7} \approx 2.65$$

4. -2.5 , $-\sqrt{5}$, $-8/3$ ✓ **Answer: $-8/3$, -2.5 , $-\sqrt{5}$**

$$-8/3 \approx -2.67, -2.5, -\sqrt{5} \approx -2.24$$